

NETWORKED CONTROL

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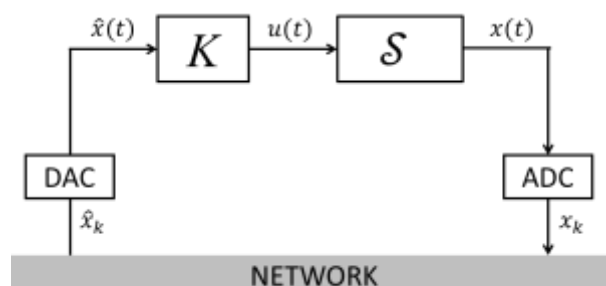
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EXERCIZES

Consider the continuous-time system

$$\dot{x} = 2x + 2u \quad (i)$$

- A) Answer the following questions concerning the continuous-time system (i).
- Compute the eigenvalue and the mode of the system. Is the system open-loop asymptotically stable?
 - Assume that the system is controlled using a continuous-time control law of the type $u(t) = Kx(t)$. Define the range of values of K that make the closed-loop system asymptotically stable.
- B) Compute the parameters F and G of the corresponding discrete-time dynamical equation
- $$x_{k+1} = Fx_k + Gu_k \quad (ii)$$
- with sampling time $h = \log_e 1.2$ s and answer the following questions concerning the discrete-time system (ii).
- Compute the eigenvalue and the mode of the system. Is the system open-loop asymptotically stable?
 - Assume that the system is controlled using a digital control law of the type $u_k = Kx_k$. Define the range of values of K that make the closed-loop system asymptotically stable.
- C) We aim to control the system using a digital quantized controller. Answer the following questions.
- According to the theory, what is the minimum rate N_B/h (where N_B is the number of bits encoding the control law and h is the sampling time) required to make the system *boundable*?
 - In view of the answer given to the previous question, what is the maximum value of h that can be used to make the system *boundable* in case $N_B = 1$, i.e., with a binary control law?
 - Set $h = \log_e 1.2$ s. Using a binary (i.e., where $N_B = 1$) control law of the type $u_k = -\text{sign}(x_k)$ analyze the properties of the state trajectories using the graphical method (possibly starting by different selected initial conditions).
- D) Assume that the system is controlled through the networked control system (NCS) sketched in the following figure.



We set $h = \log_e(\frac{12}{10})$ s and $K = -2$. Answer the following questions.

- Is the NCS asymptotically stable in case the delay is $\tau = 0$ s and in absence of packet dropouts?
- Is the NCS asymptotically stable in case the delay is $\tau = \log_e(\frac{12}{11})$ s and in absence of packet dropouts?
- Assume now that the NCS is subject to delay $\tau = \log_e(\frac{12}{11})$ s and to packet dropouts. In particular, variable θ_k has the following periodic behavior:

k	1	2	3	4	...
θ_k	1	0	1	0	...

Is the corresponding NCS asymptotically stable?

SOLUTIONS

A) The answers are given below.

- The eigenvalue of the system is $\lambda = 2 > 0$. The corresponding mode is e^{2t} . The system is open-loop unstable.
- Setting $u(t) = Kx(t)$, the closed loop dynamics is described by the equation
$$\dot{x} = (2 + 2K)x$$
whose eigenvalue is $\lambda = 2 + 2K$. The system is closed-loop asymptotically stable for $K < -1$.

B) The corresponding discrete-time system is

$$x_{k+1} = e^{2h}x_k + \frac{1}{2}(e^{2h} - 1)2u_k = 1.44x_k + 0.44u_k$$

- The eigenvalue of the system is $\lambda = 1.44$. Since $|1.44| > 1$, the system is open-loop unstable.
- Setting $u_k = Kx_k$, the closed loop dynamics is described by the equation
$$x_{k+1} = (1.44 + 0.44K)x_k$$
whose eigenvalue is $\lambda = 1.44 + 0.44K$. The system is closed-loop asymptotically stable iff $|1.44 + 0.44K| < 1$, i.e., iff $-5.5455 < K < -1$.

C) The answers are given below.

- In order to make the system *boundable*, the following inequality must be satisfied

$$\frac{N_B}{h} \geq 2 \cdot \log_2 e = 2.8854 \text{ bit/s}$$

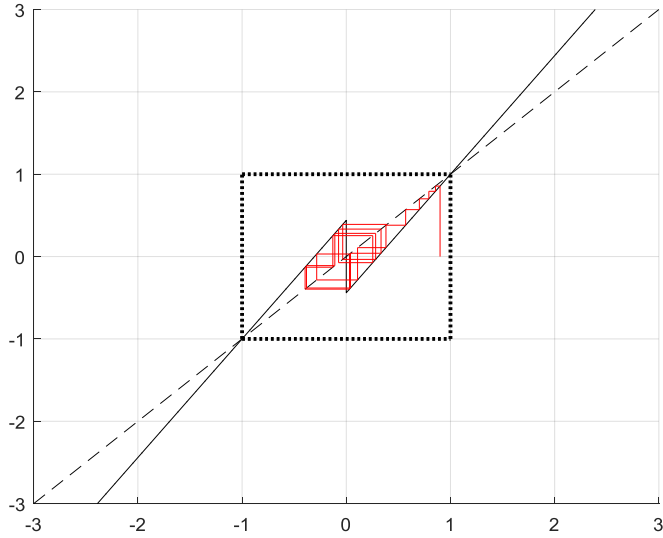
- In case $N_B = 1$, the inequality above reads

$$h \leq \frac{1}{2.8854} = 0.3466$$

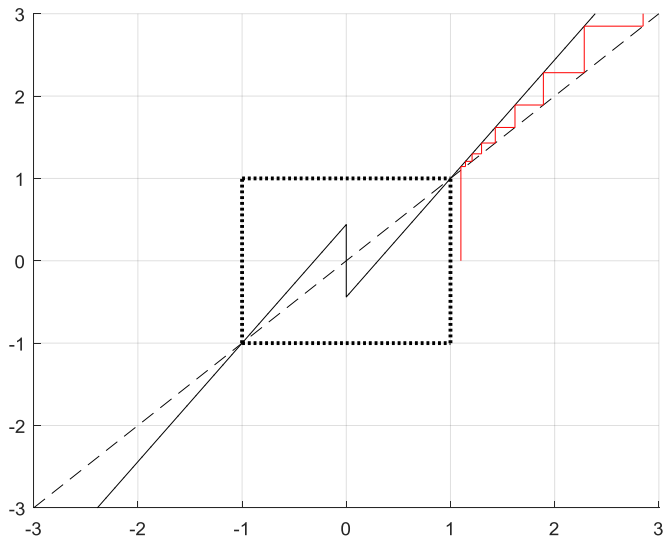
- First note that $h = \log_e(\frac{12}{10}) < 0.3466$ s. Therefore, the system can be made *boundable* using a suitable binary control law. Using the control law $u_k = -\text{sign}(x_k)$, the dynamics of the system under control is

$$\begin{cases} x_{k+1} = 1.44x_k + 0.44 & x_k \leq 0 \\ x_{k+1} = 1.44x_k - 0.44 & x_k > 0 \end{cases}$$

We apply the graphical method to analyze the properties of the state trajectories, starting from random initial conditions.



It is possible to verify that, if $-1 \leq x_0 \leq 1$, the state trajectories remain bounded in the set $[-1, 1]$. On the other hand, if $|x_0| > 1$, the trajectories diverge. This is apparent in the following plot



D) Set $h = \log_e\left(\frac{12}{10}\right)$ s and $K = -2$.

- The NCS is asymptotically stable in case delays and packet dropouts are absent, in view of the response to answer B.b.
- In case packet drops are absent, but a delay is present, the transition matrix $\Phi(h, \tau)$ is

$$\Phi(h, \tau) = \begin{bmatrix} e^{2h} + \Gamma(h - \tau)2K & e^{2(h-\tau)}\Gamma(\tau)2K \\ 1 & 0 \end{bmatrix}$$

where $\Gamma(\tau) = \int_0^\tau e^{2v} dv = \frac{1}{2}(e^{2\tau} - 1)$. In case $h = \log_e\left(\frac{12}{10}\right)$ s, $\tau = \log_e\left(\frac{12}{11}\right)$ s, and $K = -2$,

$$\Phi = \begin{bmatrix} 1.44 + 0.21K & 0.23K \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1.02 & -0.46 \\ 1 & 0 \end{bmatrix}$$

whose eigenvalues are $\lambda_1 = 0.51 + 0.4471j$ and $\lambda_2 = 0.51 - 0.4471j$, whose modulus is equal to $0.6782 < 1$. The NCS is therefore asymptotically stable.

- c. To analyze the stability of the NCS under delay and packet dropouts defined by the sequence of values of variable θ_k , we first recall that the transition matrices in case transmission occurs and in case of dropout are, respectively

$$\tilde{\Phi}_1 = \Phi(h, \tau) = \begin{bmatrix} e^{2h} + \Gamma(h - \tau)2K & e^{2(h-\tau)}\Gamma(\tau)2K \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1.02 & -0.46 \\ 1 & 0 \end{bmatrix}$$

$$\tilde{\Phi}_0 = \begin{bmatrix} e^{2h} & \Gamma(h)2K \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1.44 & -0.88 \\ 0 & 1 \end{bmatrix}$$

Note, in passing, that $\tilde{\Phi}_1$ has eigenvalues in $\lambda_1 = 0.51 + 0.4471j$ and $\lambda_2 = 0.51 - 0.4471j$, while $\tilde{\Phi}_0$ has eigenvalues in 1.44 and 1. Since the sequence of values of θ_k is 1,0,1,0,1,0,1,0:

- if k is odd, $z_{k+1} = \tilde{\Phi}_1 z_k$, $z_{k+2} = \tilde{\Phi}_0 \tilde{\Phi}_1 z_k$.
- if k is even, $z_{k+1} = \tilde{\Phi}_0 z_k$, $z_{k+2} = \tilde{\Phi}_1 \tilde{\Phi}_0 z_k$.

The update rule that can be derived is, for all $k \in \mathbb{N}$,

$$z_{2k+2} = \tilde{\Phi}_1 \tilde{\Phi}_0 z_{2k}$$

$$z_{2k+3} = \tilde{\Phi}_0 \tilde{\Phi}_1 z_{2k+1}$$

Therefore, $\tilde{\Phi}_1 \tilde{\Phi}_0$ and $\tilde{\Phi}_0 \tilde{\Phi}_1$ can be regarded as the two-steps transition matrices for the NCS state. We resort to the fact that $\tilde{\Phi}_1 \tilde{\Phi}_0$ and $\tilde{\Phi}_0 \tilde{\Phi}_1$ have the same eigenvalues to study the convergence properties of the NCS. In particular

$$\tilde{\Phi}_0 \tilde{\Phi}_1 = \begin{bmatrix} 0.5888 & -0.6624 \\ 1 & 0 \end{bmatrix}$$

whose eigenvalues are $0.2944 + 0.7588j$ and $0.2944 - 0.7588j$, whose modulus is equal to $0.8139 < 1$. The NCS is therefore asymptotically stable.